

Aircraft Price Analysis

Statistical Learning project

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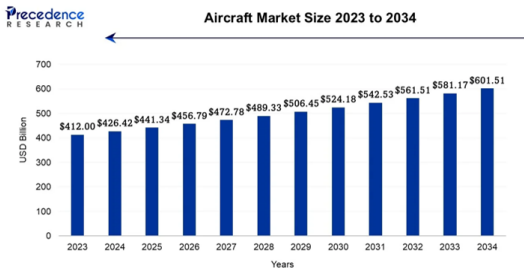
Introduction and Key Questions

Objective of the Analysis

Predict the price of aircraft using technical features and linear regression models with regularization.

Global Aircraft Market Outlook

- 2024: The global aircraft market was estimated at **USD 426.42 billion**.
- 20234: Projected to reach approximately **USD 601.51 billion**.



Source: <https://www.precedenceresearch.com/aircraft-market>

Source: Precedence Research, 2025

Aircraft Valuation: From Complexity to Innovation

- **New technologies** (e.g. electric engines, composites) make traditional valuation outdated.
- **Uncertainty** from regulations, supply chains, and market shifts adds complexity.
- **Old methods** are often slow, expensive, and subjective.
- **Industry need:** fast, data-driven models for real-time and objective pricing.

“Having a reliable data-based pricing model would be a game changer.”

— Aviation Week, 2025

Why Build a Model?

- **Aircraft value** depends on many factors: age, usage, tech specs, market demand.
- A model helps process this complexity and gives **fast, consistent estimates**.
- It reduces **subjectivity** and supports less experienced analysts.
- Once built, it can be **reused, scaled**, and improved over time
- *not a replacement for experts, but a smart step toward automation.*

Key Scientific Questions

- **Q1:** Which models achieve the highest accuracy in aircraft price prediction?
- **Q2:** Which aircraft features drive value, and how consistent are they across models?
- **Q3:** Are nonlinear effects and feature interactions relevant for pricing?
- **Q4:** Which methods offer the best trade-off between accuracy, interpretability, and efficiency?

Dataset Description

- **Total Observations:** 507
- **Total Variables:** 14
- **Numerical Variables:** 12
- **Categorical Variables:** 2
- **Target:** price (price in USD)
- **Source:** Kaggle – Aircraft Price Analysis & Prediction Dataset

Numerical Variables 1

- `engine_power`: power of the engine (HP)
- `max_speed`: maximum speed (nkots)
- `cruise_speed`: cruise speed (nkots)
- `stall_speed`: minimum speed to prevent stalling (nkots)
- `fuel_tank`: fuel tank capacity (liters)
- `range`: range (Nautical mile)
- `length`: aircraft length (Inch)

Numerical Variables 2

- `wing_span`: wingspan (Inch)
- `empty_weight`: empty weight (Pound)
- `all_eng_roc`: rate of climb with all engines (ft/min)
- `one_eng_roc`: rate of climb with one engine out (ft/min)
- `takeoff_distance`: takeoff distance (ft)
- `landing_distance`: landing distance (ft)

Categorical Variables

- `model_name`: aircraft model name
- `engine_type`: type of engine

Data Preprocessing

- Removal of rows with missing values in the target price
- Checking and handling of outliers
- Necessary transformations to normalize distributions

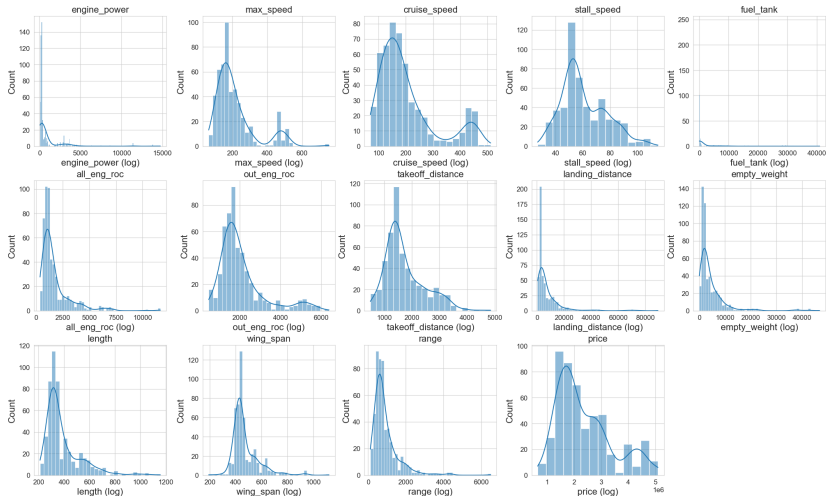
Missing Data

- 10 rows removed due to missing price
- Final dataset: 497 observations



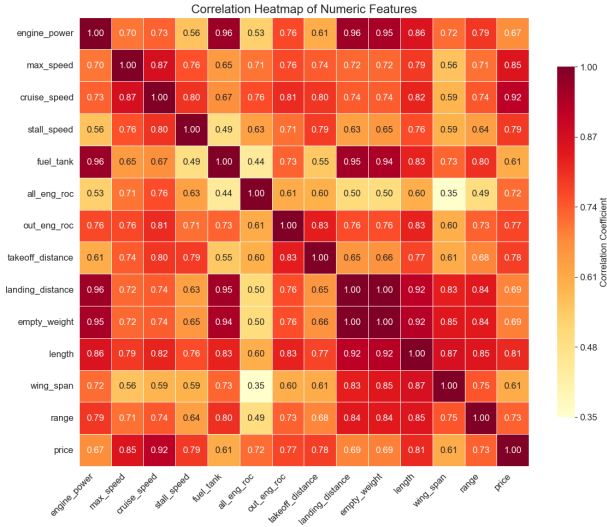
Feature Distributions

Numerical variables exhibit skewed distributions.



Correlation Matrix

Correlations between variables suggest multicollinearity.

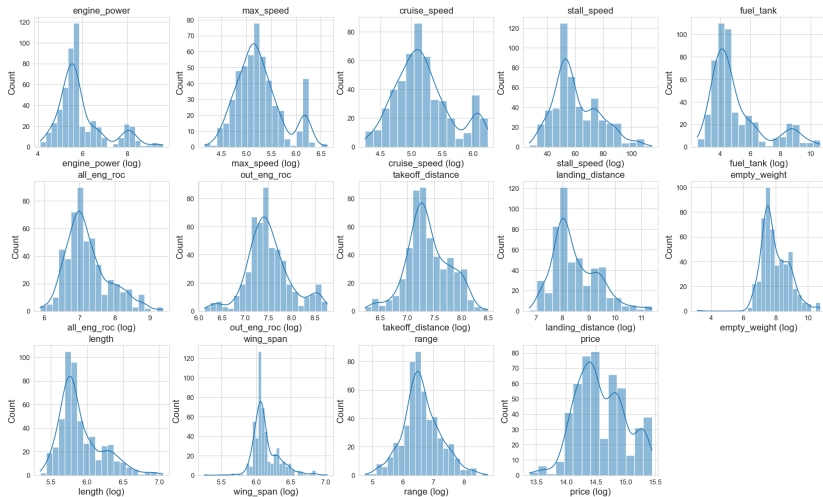


Log Transformation

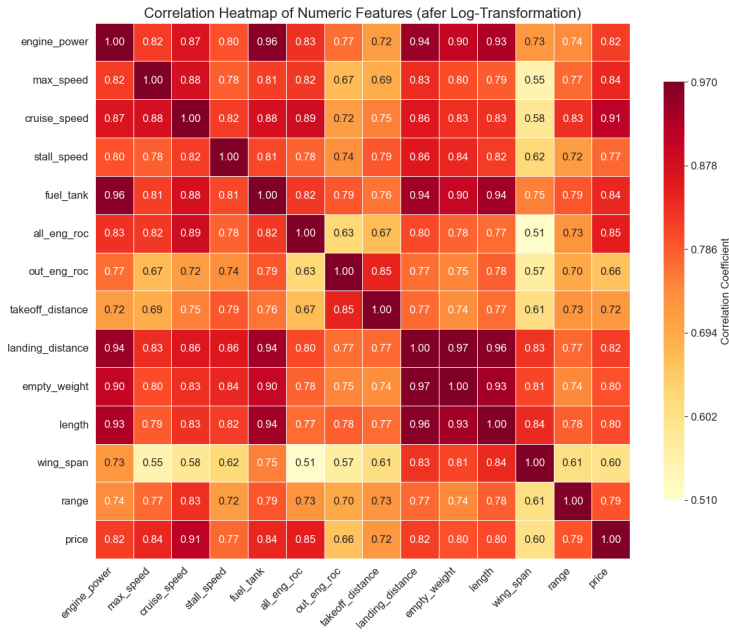
Motivations for Log Transformation

- Numerical variables have **skewed** distributions
- Logarithmic transformation applied to all numerical features to:
 - Achieve more symmetric distributions
 - Improve linearity between features
 - Reduce the impact of outliers
 - Stabilize variance

Distributions After Log Transformation



Correlation Matrix After Log Transformation



Linear Regression

Multiple Linear Regression Model

$$\log(\text{price}) = \beta_0 + \sum_{j=1}^{13} \beta_j \log(x_j + 1) + \varepsilon$$

- Preprocessing: log-transform of all numerical variables
- Validation: 10-fold CV with shuffling, `random_state=42`

K-Fold CV Metrics (Full)

=== K-Fold CV Metrics ===

R^2 mean = 0.8282

MSE mean = 0.0275

RMSE (de-logged) = 391503.27

Inferential summary aggregated over the 10 training folds is shown in the next slide.

Inferential Summary (Full)

Feature	coef _{mean}	coef _{std}	t _{mean}	p _{mean}
engine_power	0.011	0.021	0.37	0.543
max_speed	0.175	0.026	4.64	0.001
cruise_speed	0.504	0.039	9.73	0.001
stall_speed	0.001	0.001	0.87	0.430
fuel_tank	0.039	0.007	2.19	0.038
all_eng_roc	0.119	0.016	4.18	0.001
out_eng_roc	-0.099	0.006	-2.90	0.005
takeoff_distance	0.121	0.009	2.91	0.005
landing_distance	-0.325	0.059	-5.05	0.001
empty_weight	0.104	0.050	2.66	0.008
length	0.155	0.046	1.48	0.178
wing_span	0.367	0.078	3.88	0.001
range	0.025	0.011	1.06	0.336

Reduced Model (Only Significant Variables)

- Selected the 9 features with $p < 0.05$
- **Removed Features:** engine_power, stall_speed, length, range
- Retrained with the same CV settings

Reduced Model vs Full Model Results

Full Model		Reduced Model	
(14 features)		(9 features)	
R ² mean	0.8282	R ² mean	0.8291
MSE mean	0.0275	MSE mean	0.0271
RMSE (USD)	\$391,503.27	RMSE (USD)	\$389,179.25

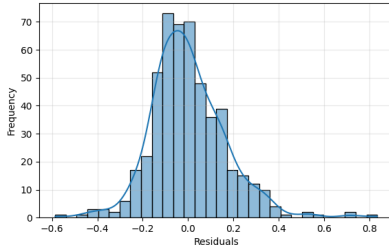
- Slight improvement in R² and RMSE

Coefficients of Reduced Model

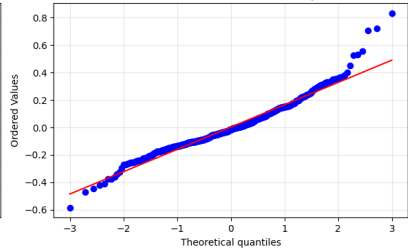
Feature	coef _{mean}	coef _{std}	t _{mean}	p _{mean}
intercept	8.515	0.311	16.37	0.001
max_speed	0.179	0.024	4.84	0.001
cruise_speed	0.522	0.032	10.81	0.001
fuel_tank	0.049	0.005	3.60	0.001
all_eng_roc	0.125	0.014	4.50	0.001
out_eng_roc	-0.084	0.007	-2.57	0.013
takeoff_distance	0.132	0.006	3.44	0.001
landing_distance	-0.285	0.059	-5.43	0.001
empty_weight	0.109	0.057	2.82	0.006
wing_span	0.405	0.057	4.75	0.001

Residual Analysis

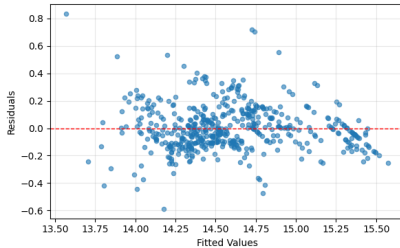
Residuals Distribution (Test Sets)



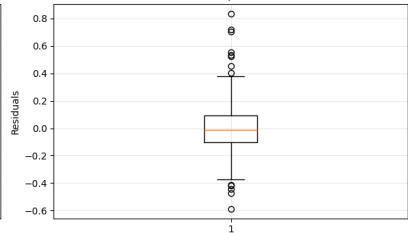
Q-Q Plot - Residuals Normality



Residuals vs Predicted Values (Test)



Residuals Boxplot (Test Sets)



Residual Analysis: Observations

- Residuals exhibit a symmetric distribution
- No obvious patterns, suggesting the model captures the variable relationships well
- A few observations have larger residuals, but no extreme outliers

Regularization

Ridge Regression

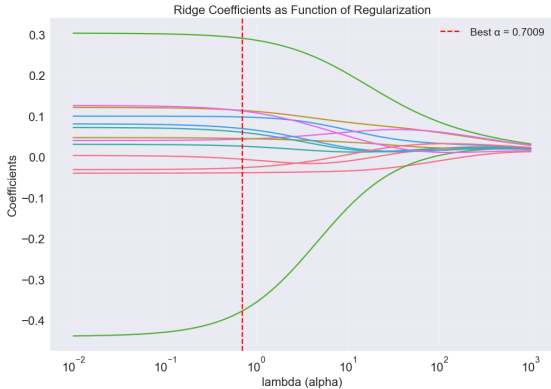
$$\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2 \right\}$$

- L2 penalty on coefficients
- Uses all features
- Cross-validation to select the best α
- Objective: reduce overfitting and improve generalization

Ridge: Metrics and Coefficients

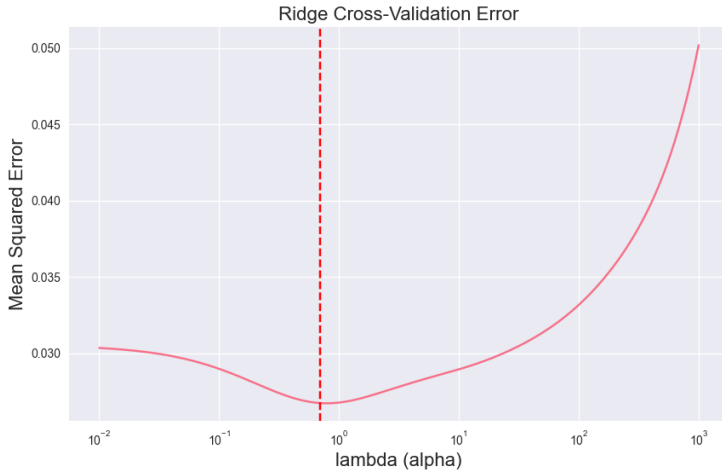
Best α and Metrics

Parameter	Value
Best α	0.7009
Test MSE	0.0265
Test RMSE	\$370,000
Test R^2	0.8495



Ridge coefficients after regularization.

Ridge: Cross-Validation Error



Cross-validation error as a function of α for selecting the optimal parameter.

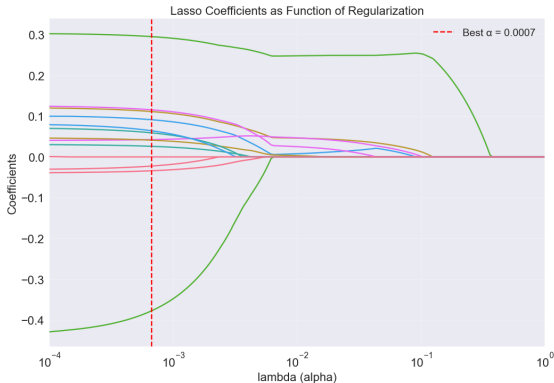
$$\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \|\beta\|_1 \right\}$$

- L1 penalty: automatic feature selection
- Coefficients can be driven to zero
- Cross-validation to select the best α
- Objective: interpretability and dimensionality reduction

Lasso: Metrics and Coefficients

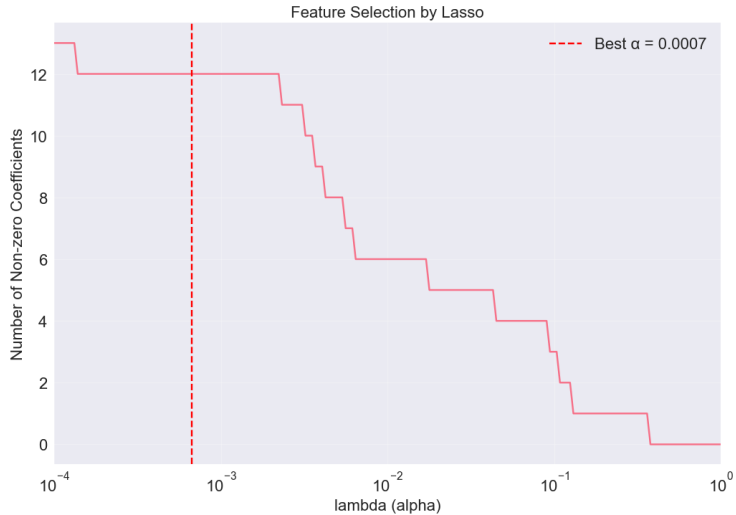
Best α and Metrics

Parameter	Value
Best α	0.0007
Test MSE	0.0265
Test RMSE	\$376,000
Test R^2	0.8494
Selected Features	12 / 13

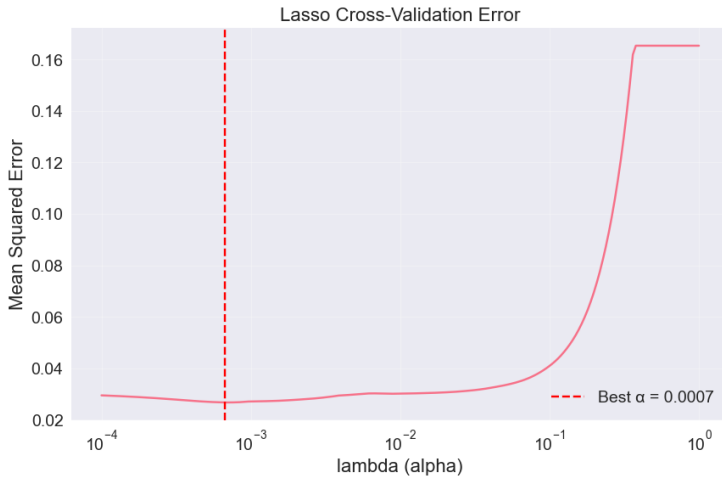


Lasso coefficients after regularization.

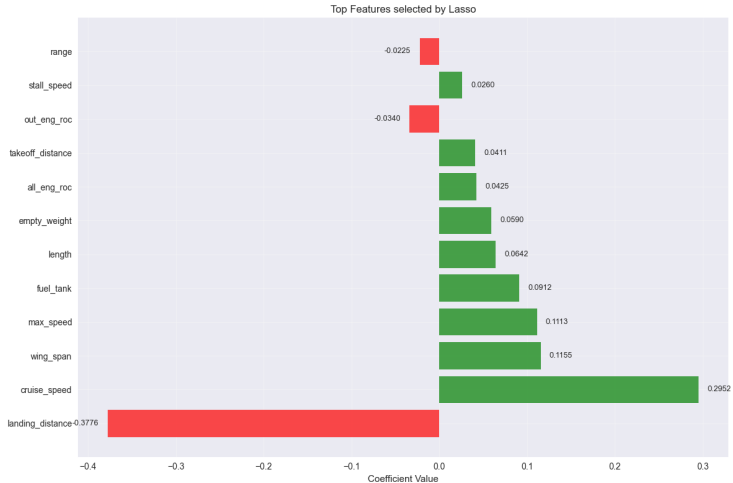
Lasso: Feature Selection



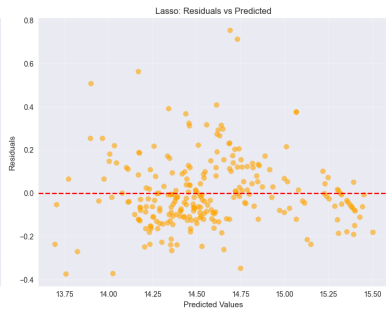
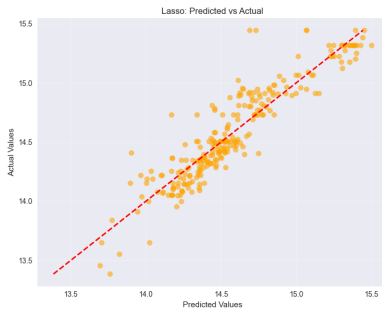
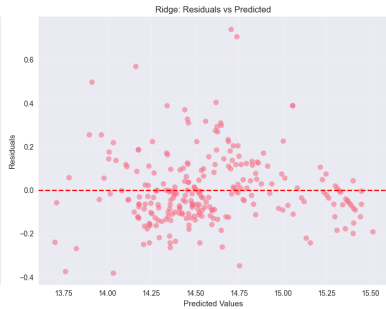
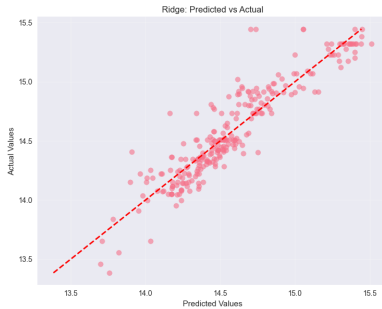
Lasso: Cross-Validation Error



Lasso: Cross-Validation Error



Residual Analysis



Residual Analysis: Observations

- Ridge and Lasso residuals show symmetry and approximate normality
- No clear patterns, suggesting both models capture relationships well
- Ridge has slightly smaller residuals, indicating better fit
- Lasso remains interpretable with fewer features and similar performance

Comparison of Linear Models

Comparison of Linear Models

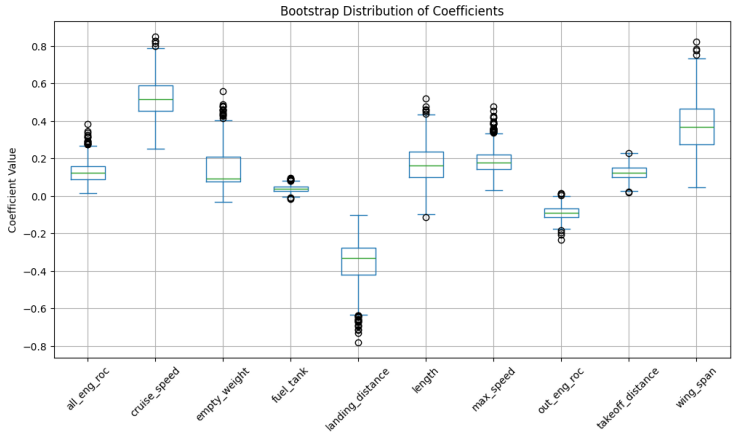
Model	R^2	RMSE (log)	RMSE (USD)
Linear Regression	0.8421	0.1668	\$389,179
Ridge	0.8495	0.1628	\$370,000
Lasso	0.8494	0.1629	\$376,000

Observations

- Ridge achieves the best R^2 (0.8584) and lowest RMSE
- Lasso remains competitive with only 16 features
- Regularization improves performance over simple linear models
- Minimal differences between Ridge and Lasso suggest robustness

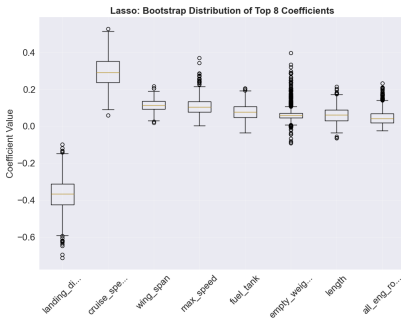
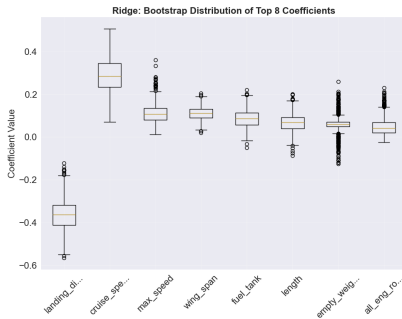
Bootstrap Inference

Linear Regression: Bootstrap Distribution of Coefficients



Boxplots of the bootstrap estimates of OLS coefficients over 2000 replications.

Ridge & Lasso: Bootstrap of Top 8 Coefficients



Comparison of bootstrap distributions for the top 8 feature coefficients in Ridge and Lasso.

Notes on Data and Coefficient Distributions

- **Estimate variability:** wider boxes indicate less stable coefficients.
- **Effect robustness:** bootstrap intervals far from zero → consistently meaningful contribution.
- **Feature selection with Lasso:** many coefficients are zero in most replications.
- **Ridge penalty:** reduces variance by shrinking coefficients toward zero without nullifying them.
- **Recommended practice:** use bootstrap to assess uncertainty and identify robust predictors.

Generalized Additive Models (GAM)

GAM Setup & Implementation

We built a multivariate GAM over the *log-price* target using all p numeric features:

- **Model form:**

$$\hat{y} = \beta_0 + \sum_{j=1}^p f_j(x_j),$$

where each f_j is a univariate smoothing spline term.

- **Terms:** one spline $s(j)$ for each of the p numeric predictors.
- **Tuning:** automatic grid-search on penalties λ_j via GCV (PyGAM's `gridsearch`).
- **Data split:** 70% train / 30% test.

GAM Final Model Evaluation

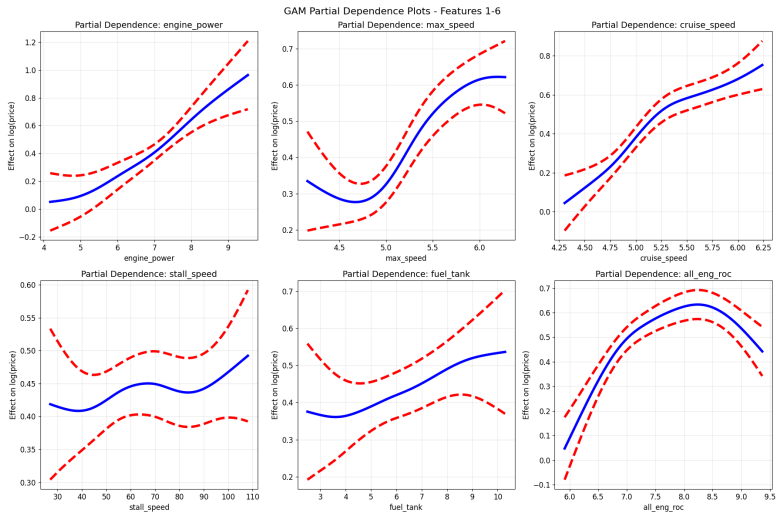
Metric	Train	Test
RMSE	0.1212	0.1455
R^2	0.9038	0.8783

Fit Statistic	Value
AIC	17 014.21
GCV	0.0204
EDF	38.09

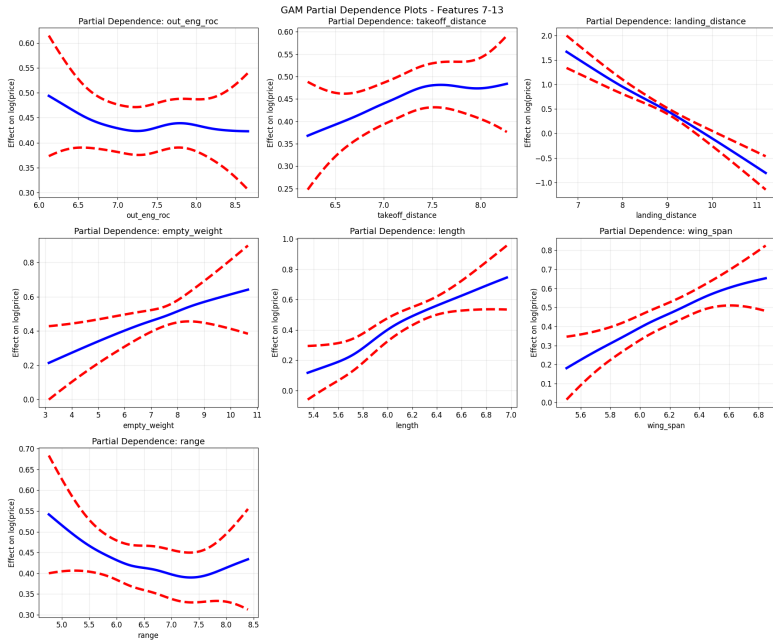
Key Takeaways

- **Small train–test gap** in both RMSE and R^2 indicates good generalization.
- **AIC & GCV** confirm the GAM's superior fit over linear and regularized benchmarks.

GAM Partial Dependence (Features 1–6)

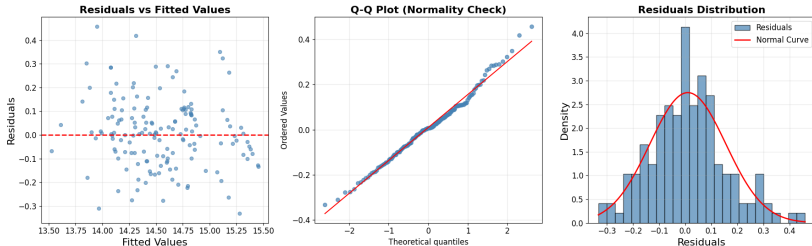


GAM Partial Dependence (Features 7–13)



Residual Analysis

GAM Model - Essential Residual Diagnostics (Test Set)



- **Homoscedasticity:** residuals are symmetrically scattered around zero with no clear pattern, indicating constant variance.
- **Approximate Normality:** Q-Q and histogram reveal a nearly Gaussian shape with only mild skew, supporting model validity.

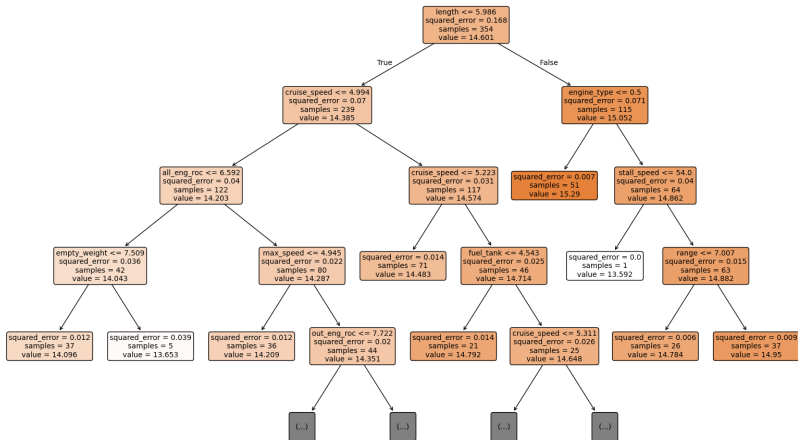
Tree-Based Methods

$$C_\alpha(T) = \sum_{m=1}^{|T|} N_m Q_m(T) + \alpha |T|$$

- Grew full tree T_0 and pruned via cost-complexity.
- Selected α^* by k-fold CV (minimizing CV RMSE).
- Explored depths $d = 1, \dots, 20$ and chose d^* via 5-fold CV.

Pruned Decision Tree (Optimal)

Best Decision Tree ($\alpha=0.000630$, depth=5)



Only first 4 levels shown for readability.

Optimal Parameters

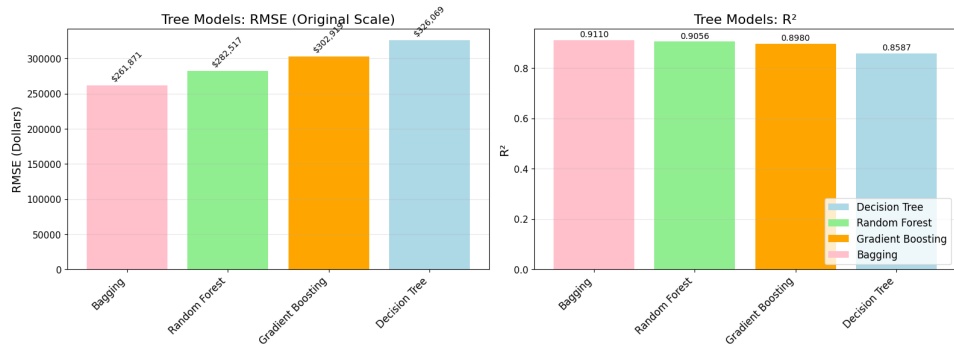
- α^* (CV): 0.00063
- Final depth: 5
- Test RMSE: 0.1567
- Test R^2 : 0.8587

Pruning reduces overfitting while preserving accuracy.

After pruning the decision tree, we implemented and compared three ensemble methods:

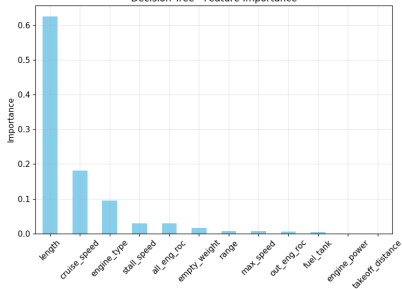
- Random Forest
- Gradient Boosting
- Bagging

Model Performance Comparison

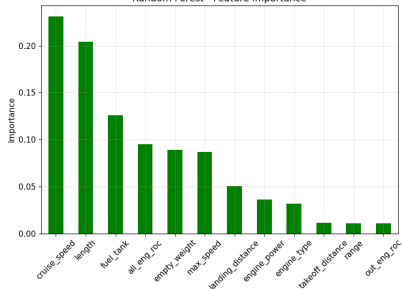


Feature Importance

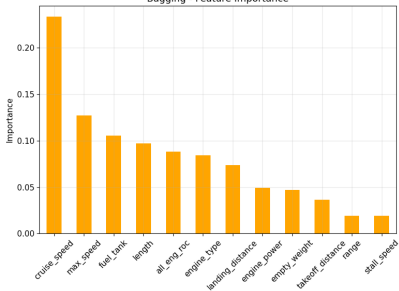
Decision Tree - Feature Importance



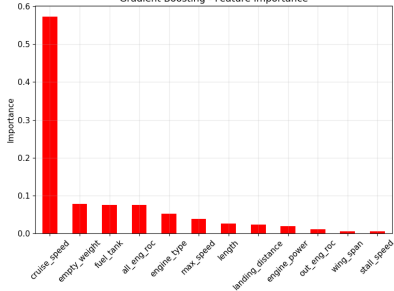
Random Forest - Feature Importance



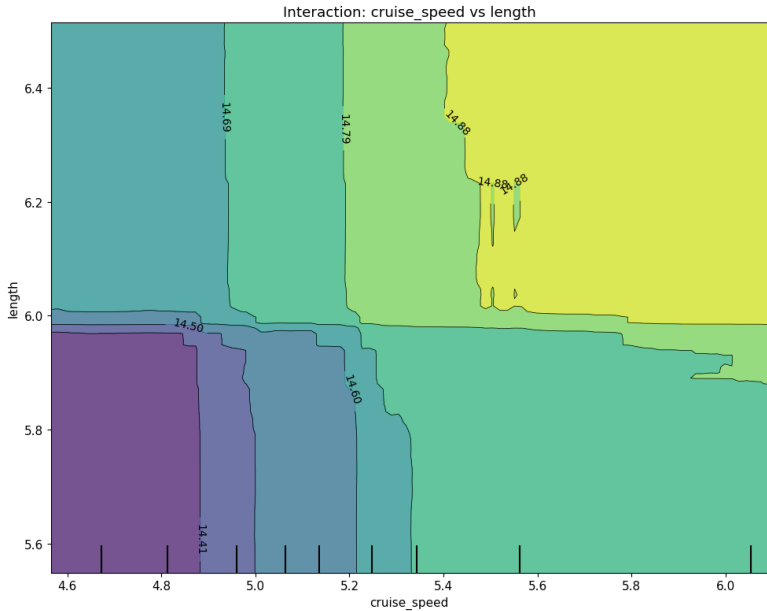
Bagging - Feature Importance



Gradient Boosting - Feature Importance

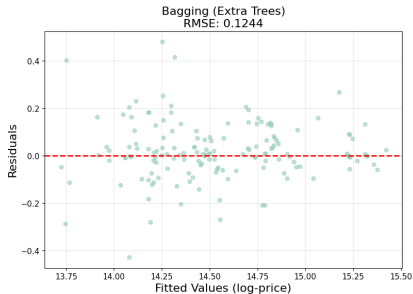
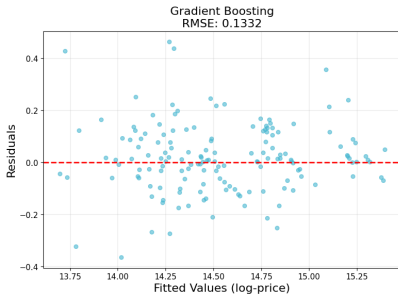
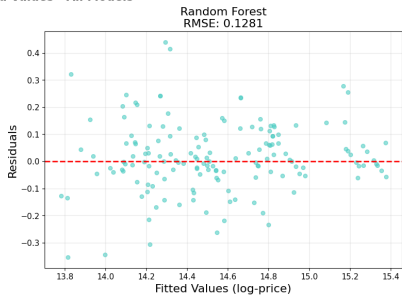
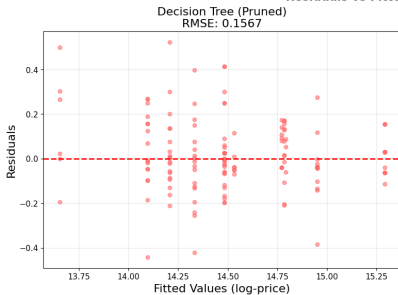


RF Partial Dependence: Cruise Speed $\times$ Length



Residuals vs Fitted

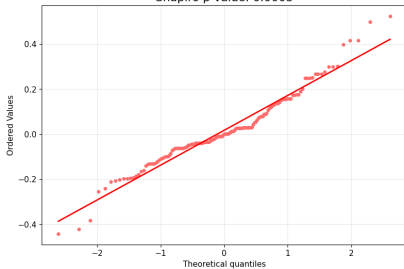
Residuals vs Fitted Values - All Models



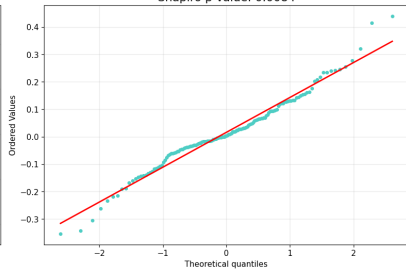
Q-Q Plot of Residuals

Q-Q Plots - Normality Test of Residuals

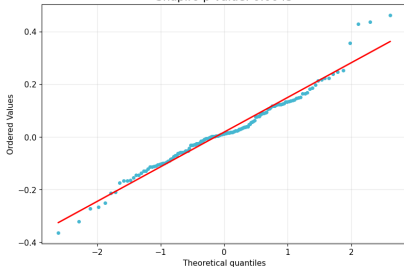
Decision Tree (Pruned)
Shapiro p-value: 0.0003



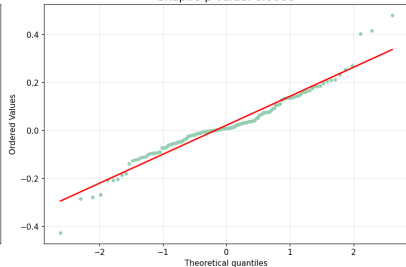
Random Forest
Shapiro p-value: 0.0084



Gradient Boosting
Shapiro p-value: 0.0045



Bagging (Extra Trees)
Shapiro p-value: 0.0000



- **Homoscedasticity:** residual spread is uniform, no clear pattern.
- **Normality:** slight deviations from normality, but acceptable.
- **Residual variance:** lowest for Random Forest, indicating best fit.

Conclusions

Performance Summary

Model	RMSE (USD)	R ²
Linear Regression	\$389,179	0.8421
Ridge	\$370,000	0.8495
Lasso	\$376,000	0.8494
GAM	\$361,000	0.8920
Pruned Decision Tree	\$326,069	0.8587
Random Forest	\$282,517	0.9056
Gradient Boosting	\$302,919	0.8980
Bagging (Extra Trees)	\$261,871	0.9110

Which One Is The Best Model ?

Top Performer

- **Bagging (Extra Trees)** achieves the *lowest RMSE* (\$261,871) and *highest R^2* (0.9110).

Interpretability Trade-off

- **Bagging/Ensembles:** – *High predictive power*, but *low interpretability*. – Harder to extract feature-level insights.
- **GAM / Linear Models:** – Slightly lower accuracy ($R^2 \approx 0.89$), but *high interpretability*. – Coefficients/partial-dependence plots give clear effects.
- **Tree-based Models:** – Pruned Decision Tree offers *some interpretability* but lower performance. – Random Forest sits between in both accuracy and explainability.

Acknowledgments & References

- **Book:** James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning: with Applications in Python*. Springer Cham.
- **Course:** “Statistical Learning” at University of Bergamo, Prof. Francesco Finazzi
- **Exercises:** Python notebook exercises developed by Prof. Francesco Finazzi for the UnivBG course

Thank you for your attention!

Questions?